

Course Title	Digital Logic Design Lab	Course Code	EL1005
Department	Department of Electrical Engineering (DEE)	Campus	Lahore
Knowledge Profile	Engineering Design (WK5)	Credit Hrs.	1
Knowledge Area	Computer Engineering (KA01)	Grading Scheme	Relative
HEC Knowledge Area	Engineering Foundation	Applicable From	Fall 2023
Pre-requisite(s)			

Course Objective	The objective of this lab is to implement the design and analysis of basic digital logic, combinational and sequential circuits; using random logic gates, decoder/encoder, multiplexers, flip flops, registers and counters.
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No.	Assigned Program Learning Outcome (PLO)
3	An ability to design solutions for complex engineering problems and design systems, components or processes that meet specified needs with appropriate consideration for public health and safety, cultural, societal, and environmental considerations.
4	An ability to investigate complex engineering problems in a methodical way including literature survey, design and conduct of experiments, analysis and interpretation of experimental data, and synthesis of information to derive valid conclusions.
5	An ability to create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, to complex engineering activities, with an understanding of the limitations.
9	An ability to work effectively, as an individual or in a team, on multifaceted and /or multidisciplinary settings

CP = Class Participation, L= Lab, PR = Project Report, PQ = Project Quiz, Q = Quiz, PD = Project Demonstration.

No.	Lab Learning Outcome (LLO) Statements	Assessment Tools	Taxonomy Levels	PLO
1	Display active individual / teamwork and high ethical standards.	CP1-CP12	A5	9
2	Practice experiments under supervision to acquire the required data/results using modern tools.	L1-L12	P3	5
3	Apply the knowledge of subject in the lab environment.	PR, PQ, Q1	C3	4
4	Build a product or system meeting the requirements.	PD, Q2	P5	3

Text Books	Title	
	Author	
	Publisher	
Reference Books	Title	
	Author	
	Publisher	
	Title	
	Author	
	Publisher	

Week	Course Contents/Topics	Chapter*	LLO
01	Introduction to logic trainer and logic probe		1,2
02	Basic Gates & trouble shooting		1,2
03	Boolean Algebra		1,2
04	K-Maps		1,2
05	Universal Gates Mapping		1,2
06	Binary Comparator		1,2
07	Decoder, Encoders, Multiplexers		1,2
08	Implementation of logic circuit using Decoder & MUX IC'S		1,2
09	Adder & Subtractor		1,2
10	Introduction to logic works		1,2
11	Design & Analysis of Sequential Circuits		1,2
12	Registers		1,2
13	Project Week		3,4
14	Project Week		3,4

Assessment Tools	Weightage
Class Participation	10%
Lab Work	40%
Quiz	20%
Project (Quiz, Report, Demonstration)	5%+5%+20%